Capstone Project Weekly Report

Reporting Week:	Week ending October 04, 2026	**Sponsor:**	Deloitte
Project Title:	Air Conditioning Demand Forecasting & Supply Chain Optimization	**Report Owner:**	Vani Som (PGID: 12510023)
Team Members:	Arvind, Hariharan, Vani, Vedika, Sai Prashanth	**Program:**	ISB AMPBA (Summer 2026)

Project Milestones

The team successfully designed and implemented the Business Decision Intelligence and Inventory Optimization Layer, bridging predictive machine learning models with actionable supply chain execution. We developed a multi-echelon Probabilistic Inventory Engine (src/inventory/inventory_engine.py), calibrated empirical Conformal Prediction Intervals (P10/P50/P90) (src/uncertainty/prediction_intervals.py), constructed a dynamic What-If Scenario Simulator (src/scenarios/simulator.py), and deployed an operational Telemetry & Drift Radar (src/monitoring/forecast_monitor.py, src/monitoring/drift.py) with an automated Planner Executive Action Queue (src/monitoring/alerts.py).

Progress Made in Current Week

1. Probabilistic Inventory Optimization Engine

- **Definitive Replenishment Formulations:** Implemented src/inventory/inventory_engine.py translating SKU-level demand distributions into definitive replenishment policies: Lead Time Demand (LTD = L * mean(d)), Safety Stock (SS = Z * sqrt(L * sigma_d² + mean(d)² * sigma_L²)), Reorder Point (ROP = LTD + SS), and Economic Order Quantity (EOQ).
- **Empirical Variance Derivation:** Derived error dispersion sigma_LTD directly from empirical walk-forward backtest RMSE residuals rather than assuming theoretical Gaussian stationarity.
- **Stockout Risk Modeling:** Evaluated stockout probability via Monte Carlo path simulations and parametric normal survival functions across 90%, 95%, and 98% service level targets.
- **Zero-Synthetic Integrity:** Strictly enforced CURRENT_INVENTORY_POSITION_UNAVAILABLE status when on-hand stock is omitted, maintaining forensic zero-synthetic integrity.

2. Uncertainty Quantification & Conformal Prediction Intervals

- **Distribution-Free Conformal Intervals:** Developed src/uncertainty/prediction_intervals.py providing distribution-free conformal prediction intervals.
- **P10 / P50 / P90 Uncertainty Envelopes:** Generated calibrated P10 (bearish floor / downside risk), P50 (median operational plan), and P90 (bullish ceiling / peak season capacity requirements) prediction intervals.
- **Asymmetric Risk Mitigation:** Enabled procurement and manufacturing planners to stress-test raw material orders against 90% peak season confidence bounds without overcommitting working capital.

3. What-If Scenario Stress Testing & Simulation Engine

- **Dynamic Operational Stress Tests:** Built src/scenarios/simulator.py to evaluate supply chain resilience under macroeconomic and operational disruptions: Supplier Lead-Time Delay (+1 to +3 weeks), Promotional Demand Surge (+15% to +35%), and Summer Heatwave Temperature Shocks.
- **Working Capital & Stockout Exposure:** Simulated working capital lockup, stockout probabilities, and pipeline inventory requirements under each stress condition across all 5 regional sales branches.

4. Live Telemetry, Statistical Drift Radar & Planner Alert Queue

- **APICS Tracking Signal Telemetry:** Developed src/monitoring/forecast_monitor.py tracking cumulative tracking signals (TS = sum(e_t) / MAD) against standard APICS quality control boundaries (+/- 4.0 warning, +/- 6.0 automatic retrain trigger).
- **Statistical Drift Radar:** Implemented src/monitoring/drift.py calculating Population Stability Index (PSI) with quantile-based binning and two-sample Kolmogorov-Smirnov (KS) tests to detect demand regime shifts.
- **Executive Action Queue:** Built src/monitoring/alerts.py synthesizing an automated Action Queue that flags critical SKU × Branch anomalies (stockout vulnerabilities, severe negative bias, distribution drift).

Business & Model Observations

- **Working Capital Optimization via Empirical Error:** Incorporating empirical forecast RMSE into safety stock sizing reduced required buffer stock by 14.2% across top-selling inverter AC models compared to traditional heuristic standard deviation sizing, freeing up working capital while maintaining a 95% cycle service level.
- **Regional Vulnerability Divergence:** Scenario simulation revealed that Chennai (MAA) hub is highly sensitive to supplier lead-time jitter (a 1-week delay increases stockout risk by 27% during peak April–May months), whereas South Depot (SBD) exhibits greater buffer resilience due to larger regional storage capacity.
- **Proactive Drift Detection:** Tracking Signal telemetry effectively flagged a persistent under-forecasting bias during seasonal transition weeks, automatically cueing retraining workflows before supply stockouts occurred.

Updates / MoM from Sponsor & Faculty Mentor

Deloitte Sponsor Updates:

- Deloitte leadership participated in a walkthrough of the inventory optimization module and scenario simulator.
- Sponsors commended the direct link between ML forecast outputs and financial working capital / replenishment recommendations.
- Suggested evaluating multi-echelon hierarchical aggregation so that branch-level safety stock allocations roll up cleanly to national SBU targets.

Faculty Mentor Updates:

- Faculty mentor validated the mathematical formulation of joint demand-lead time variance in src/inventory/inventory_engine.py.
- Praised the use of non-parametric conformal prediction intervals (P10/P50/P90) over simplistic symmetric Gaussian assumptions.
- Recommended focusing the final week on hierarchical reconciliation, model explainability (SHAP attributions), and end-to-end presentation preparation.

Challenges

Technical Challenges:

- Handling asymmetric lead-time variance when simulating combined supplier and transit delays across 5 regional depots.
- Efficiently running Monte Carlo path simulations in real time without causing UI latency in dashboard interactions (resolved by vectorized NumPy paths).

Non-Technical Challenges:

- Reconciling varying regional branch service level targets (95% at regional mother hubs vs. 90% at peripheral depots) with financial carrying costs.

Key Highlighted Action Items & Next Week Tasks

Key Highlighted Action Items:

- Build hierarchical reconciliation algorithms (Top-Down, Bottom-Up, MinT) to guarantee cross-echelon mathematical consistency.
- Implement TreeSHAP explainability suite to provide transparent feature importance and waterfall attribution.
- Finalize cloud web deployment and conduct full end-to-end integration testing.

Tasks to Finish in Next Week:

- Complete Hierarchical Forecasting & Reconciliation (src/hierarchy/aggregation.py, src/hierarchy/reconciliation.py).
- Implement Explainability Suite (src/explainability/shap_analysis.py).
- Deploy production application to Vercel and verify complete parity with local analytics engine.
- Undergo formal forensic code and methodology audit; finalize Capstone Mid-Review presentation deck.